

Algebra 1
Unit 1
Lesson 10 Practice

Name _____
Date _____
Period _____

1. To multiply two _____ with the same index, multiply the integers together and then multiply the radicands together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

2. $\sqrt{11} \cdot \sqrt{11}$

3. $\frac{3\sqrt{3}}{-\sqrt{48}}$

4. $\sqrt[3]{8} \cdot \sqrt[3]{16}$

5. $\sqrt{10} \cdot \sqrt{6}$

6. $\frac{\sqrt[3]{432}}{\sqrt[3]{128}}$



7. $\sqrt[3]{98} \cdot \sqrt[3]{21}$

8. Identify all equivalent expressions to $\sqrt{7} \cdot \sqrt{14}$.

- $\sqrt{98}$
- $7\sqrt{7}$
- $(7 \cdot 14)^{\frac{1}{2}}$
- $7\sqrt{2}$
- $14\sqrt{2}$
- $7^{\frac{1}{2}} \cdot 14^{\frac{1}{2}}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{-5\sqrt{3}}{2\sqrt{5}}$

10. $\frac{\sqrt{27}}{3\sqrt{6}}$